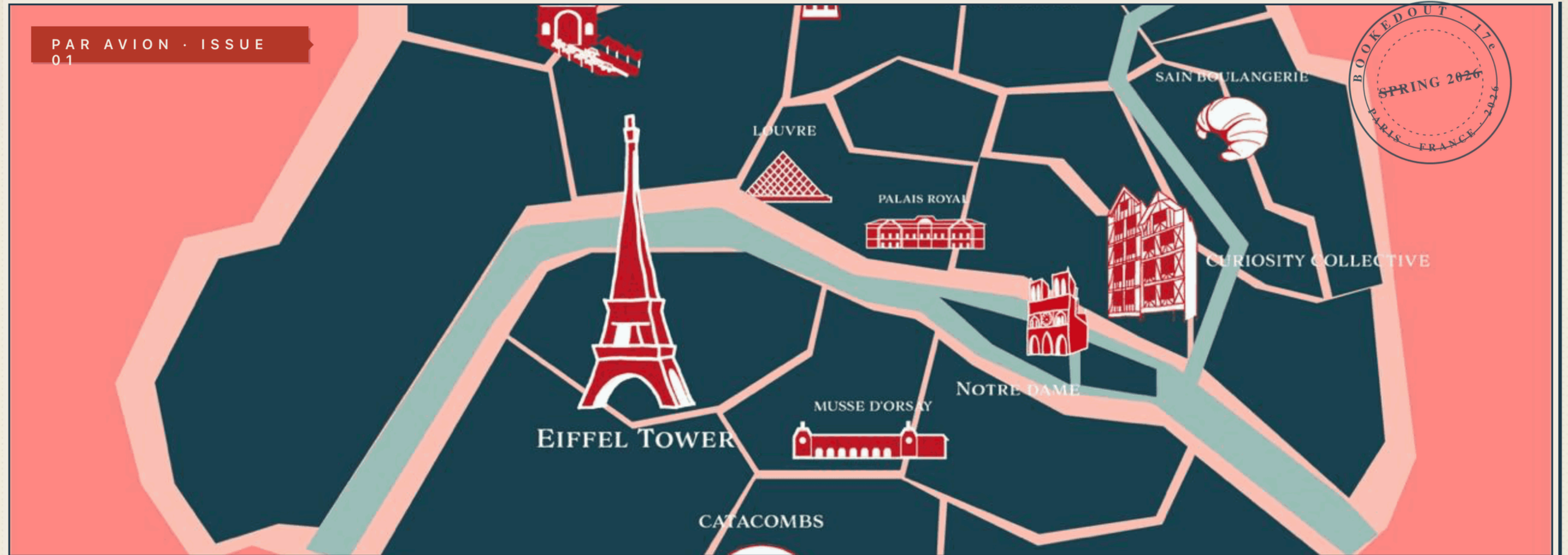
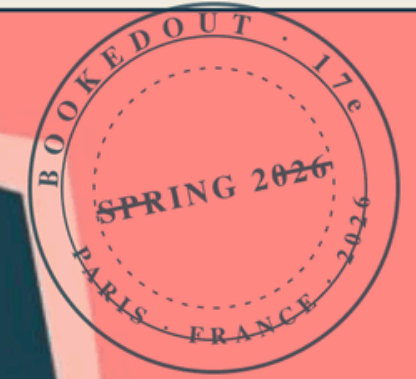


PAR AVION · ISSUE
01



BookedOut.

TOPIC

Airbnb Market Analysis
Batignolles-Monceau · 17e
Paris · France

PROJECT
AIRBNB PARIS MARKET

PRESENTED BY
PRINCE CHINYAMA
KAMALESH KUMAR

DATE
2026 · 05

SHEET

PART I · POSTCARD FROM KAGGLE

Predictive Modelling.

We started with nineteen variables and a logistic regression and finished with ninety-three features. The jump from 0.781 to 0.861 came from feature engineering, not algorithm selection, and the equal blend of LightGBM, XGBoost, and CatBoost beat any single learner and the optimized stack.

METHOD	Kaggle competition
TRAINING ROWS	374,796
FEATURES	95
SUBMISSIONS	47 documented



PUBLIC LEADERBOARD · FINAL

#2

TWO RANKS FROM FIRST · 0.861 AUC

AUC LIFT

+10.1%

Baseline 0.781 → final 0.861

SUBMISSIONS

47

Daily iteration, logged

Results at a glance.

Public leaderboard, final standing. Documented across daily iterations.

R · 01

PARIS

#2

Final standing, held-out test set.

PUBLIC LEADERBOARD

R · 02

PARIS

0.861

Confirmed via 5-fold cross validation.

BEST PUBLIC SCORE

R · 03

PARIS

47

Each documented in submission log.

SUBMISSIONS · DAILY

R · 04

PARIS

+10.1%

Lift attributable to feature engineering.

AUC LIFT VS. BASELINE

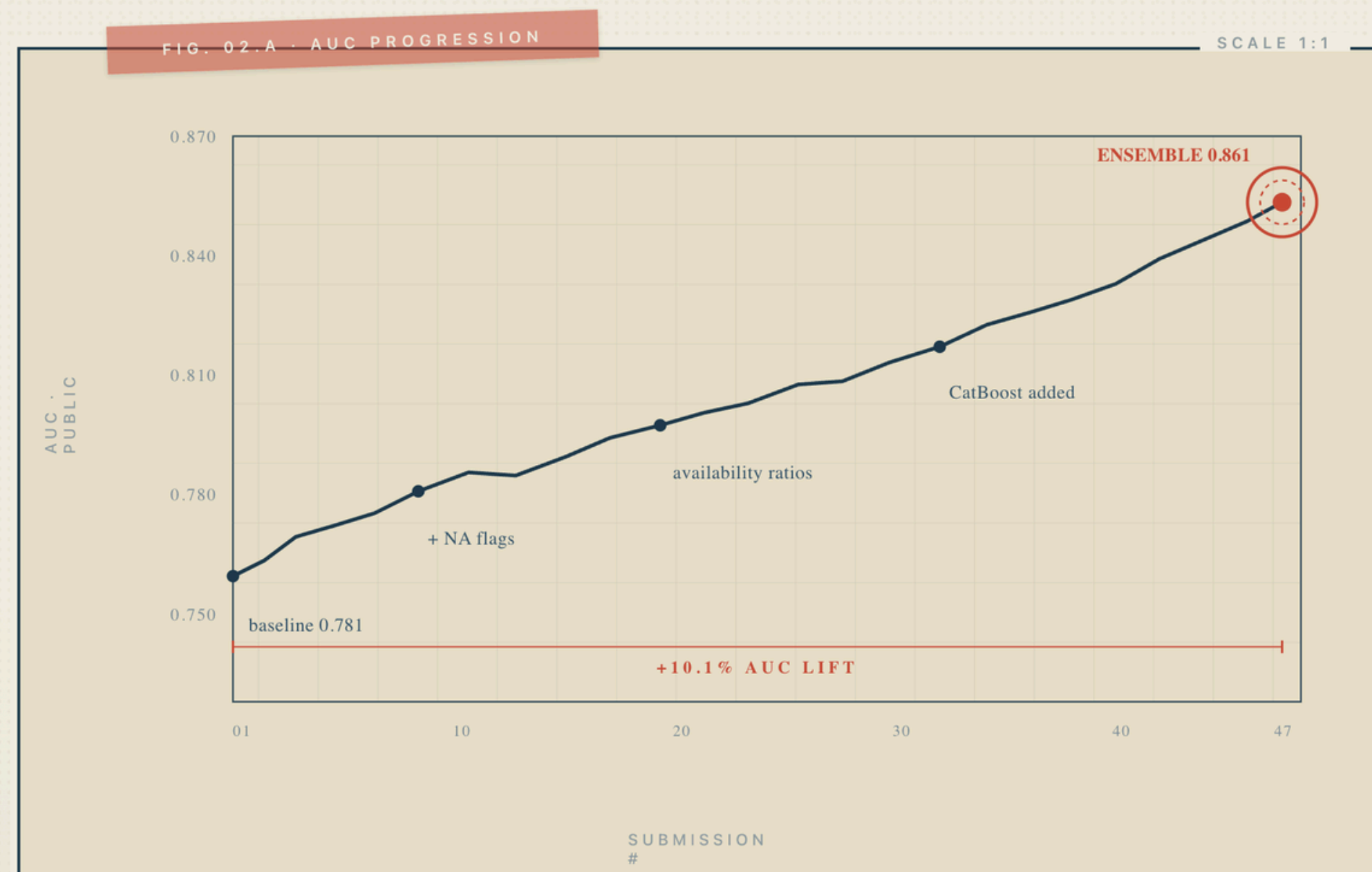
GENERAL NOTE

Iteration speed and feature thinking mattered more than algorithm selection. The same three variables that drove Kaggle gains are exactly what Batignolles-Monceau rewards on the ground.

PART I · MODELLING	DRAWING P-04 SCHEDULE	SCALE N.T.S.	SHEET
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From baseline to ensemble. AUC across 47 submissions.

Feature engineering, not algorithm selection, produced the lift from 0.781 to 0.861.



N · 01

Feature engineering drove the biggest gains.

NA flags, availability ratios, and review-to-tenure unlocked five of the six largest single-submission jumps.

N · 02

CV AUC tracked public scores throughout.

Genuine generalisation, not leaderboard overfitting. The gap held at plus or minus 0.004 across 47 runs.

N · 03

LGB + XGB + CatBoost ensemble.

The ensemble outperformed any single learner by 0.012 AUC. Daily iteration, 47 documented submissions.

Feature engineering, not algorithm selection.

The same three variables that built the leaderboard model are the ones driving bookings on the ground.

ZONE A · KAGGLE

Three features produced the largest AUC gains.

F01 Price relative to market
Log price vs. neighbourhood median.

F02 Distance from tourist zone
Haversine distance to the Eiffel centroid.

F03 Cancellation flexibility
Flexible · moderate · strict.



SAME LOGIC

ZONE B · BATIGNOLLES-MONCEAU

The same variables drive bookings on the ground.

Q2 Price
OR 0.593 per log unit, $p < 0.001$. Strongest coefficient.

Q1 Location
OR 1.125 per km from Eiffel, controlling for price.

Q3 Cancellation
+12.4 pp causal ATT vs. matched strict policy.

PART
I-II BRIDGE

DRAWING
P-03 BRIDGE

SCALE
N.T.S.

SHEET



PART II · POSTCARD FROM PARIS

Market Analysis.

Three research questions, three methods, one investor recommendation at the end.

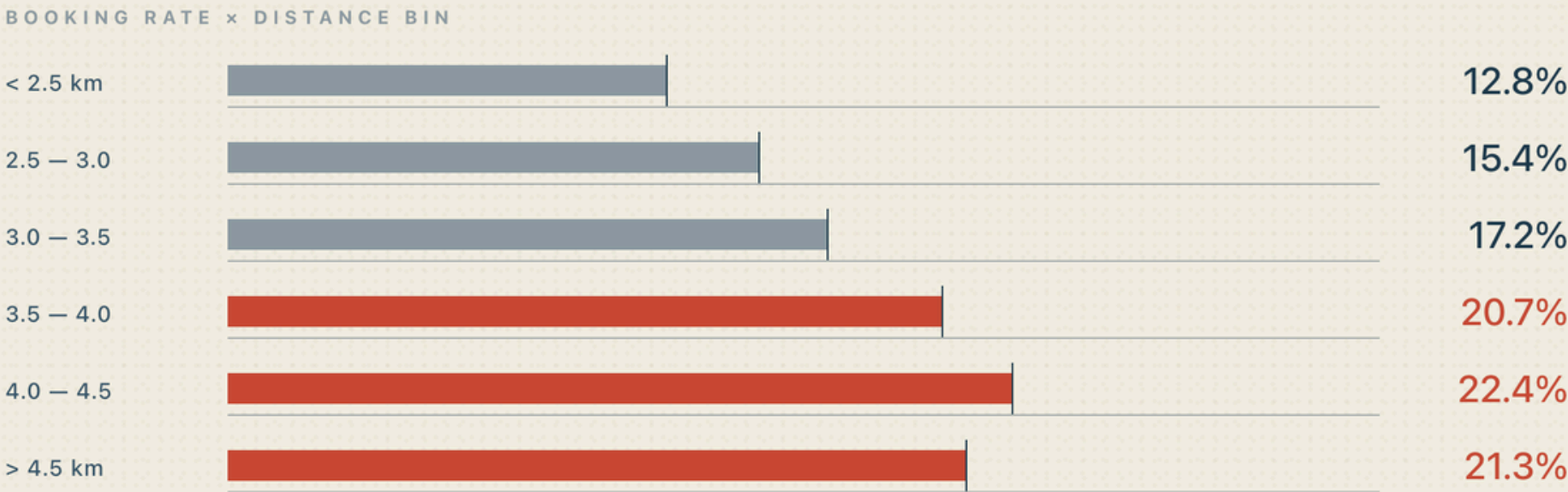
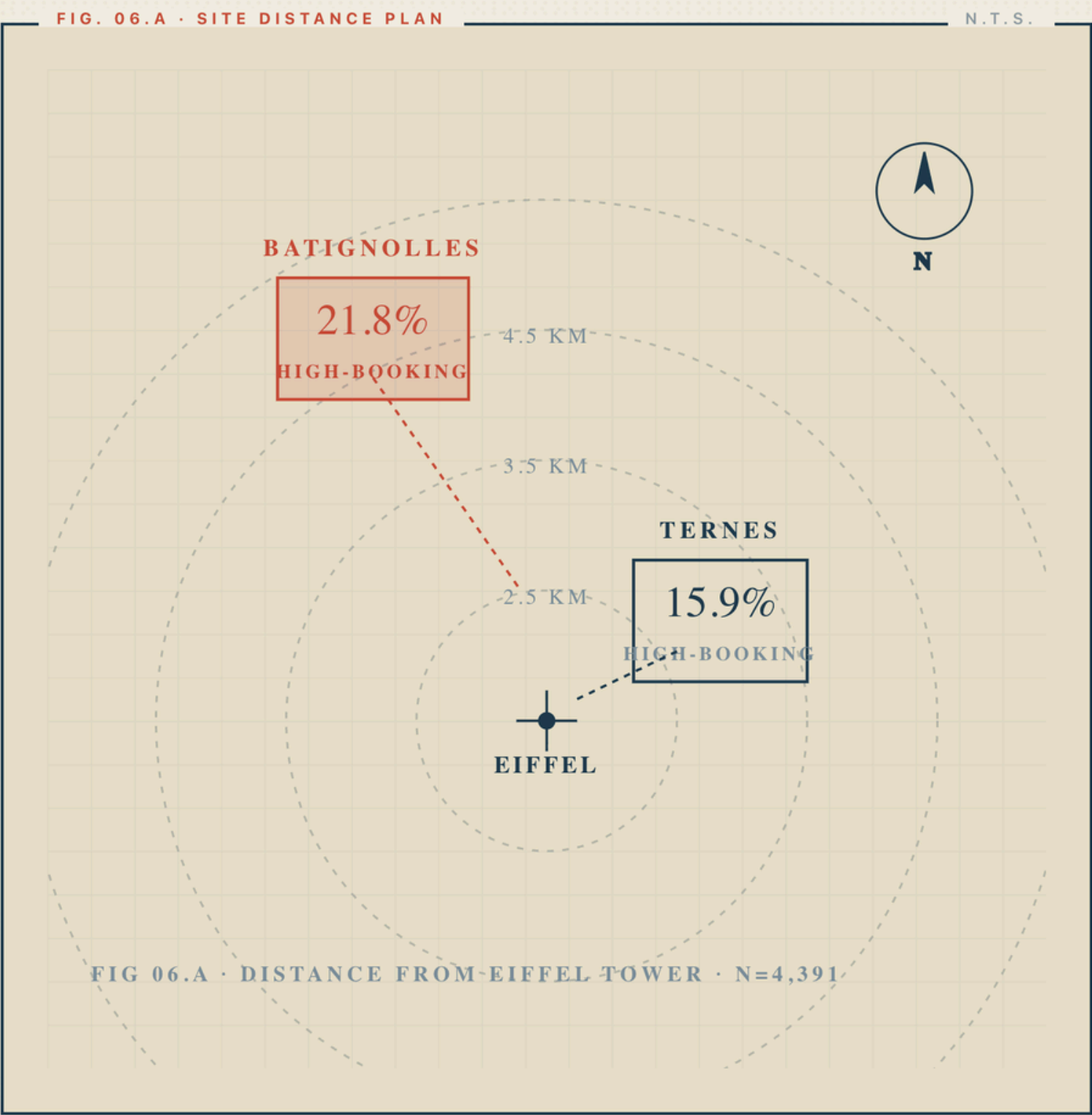
SITE	Batignolles-Monceau · 17e
LISTINGS	4,391 unique
METHODS	Logit · PSM · XGBoost
OUTPUT	Investor recommendation

RESEARCH AGENDA · Q1 · Q2 · Q3

NO.	CATEGORY	RESEARCH QUESTION
Q1	LOCATION	Does north-south split predict booking success independently of price?
Q2	PRICE	What price maximises bookings under the 120-night Paris cap?
Q3	OPERATIONS	Which post-acquisition decisions actually move booking rates?

Location matters independently of price. Buy north.

OR 1.125 per km from Eiffel Tower, controlling for price. The northern Batignolles cluster outperforms the tourist-adjacent south.

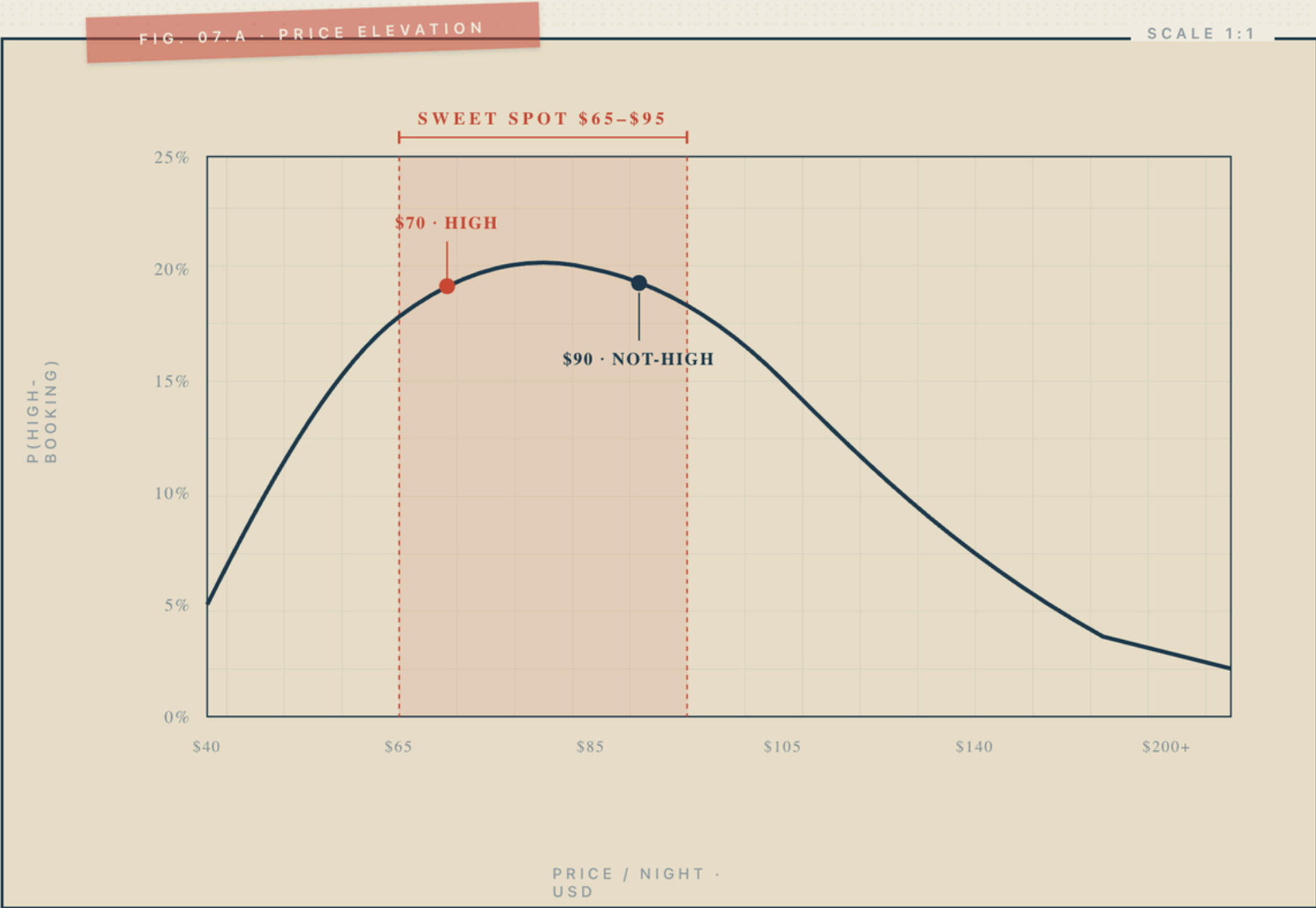


◆ North · Batignolles		◆ South · Ternes	
LISTINGS	2,242	LISTINGS	2,149
HIGH-BOOKING	21.8%	HIGH-BOOKING	15.9%
MEDIAN PRICE	\$75	MEDIAN PRICE	\$99
FLEX CANCEL	54.6%	FLEX CANCEL	49.6%

PART II · Q1	DRAWING P-06 SITE PLAN	SCALE 1:50,000	SHEET
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Price is the dominant driver. Sweet spot at \$65 to \$95 per night.

The strongest coefficient in the entire model. Above \$100 a listing exits the sweet spot entirely.



4,391

UNIQUE LISTINGS

18.9%

HIGH-BOOKING RATE

\$85

MEDIAN PRICE / NIGHT

3.7%

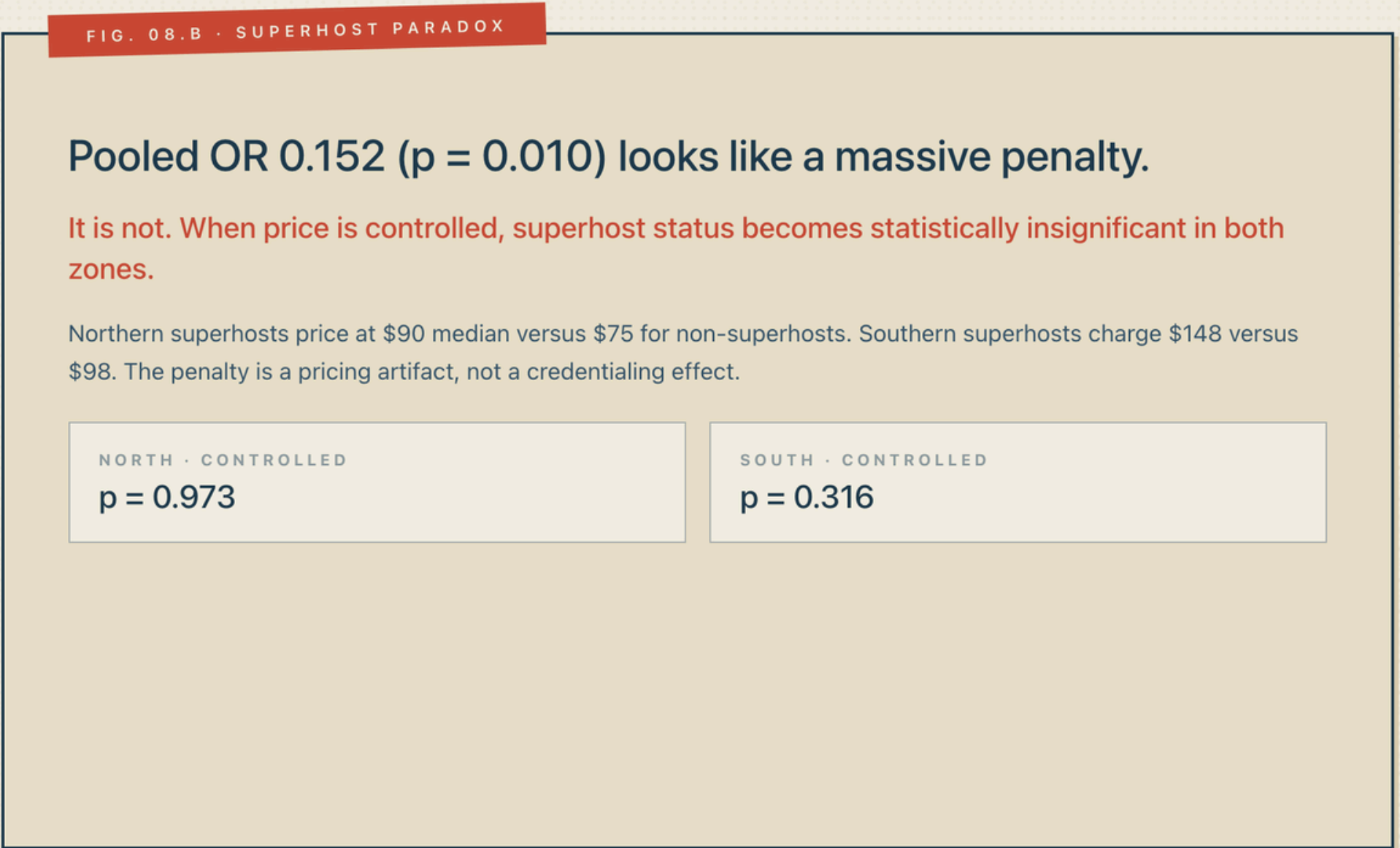
SUPERHOST RATE

KEY FINDING

Above \$100 per night a listing does not pay a marginal booking cost. It exits the sweet spot entirely.

Flexible cancellation works in the north. Superhost status doesn't move bookings.

Propensity-score matched analysis. When price is controlled, the apparent superhost penalty disappears.



PART II · Q3	DRAWING P-08 DETAILS	SCALE N.T.S.	SHEET
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XGBoost validates the regression. Top 10 listings share one profile.

Three methods converge on the same property: northern, flexible, entire-home, \$25 to \$70 per night.

SHARED PROFILE · XGBOOST TOP 10

KEY	ATTRIBUTE	SPECIFICATION
N	Northern zone	3.8 to 4.4 km from Eiffel
H	Entire home or apartment	10 of 10 listings
F	Flexible cancellation	9 of 10 listings
\$	Inside the sweet spot	\$25 to \$70 / night
P	Predicted P(high-booking)	49% to 59%

◆ Regression. PSM. XGBoost. Three methods, one profile.

TOP 10 LISTINGS · PREDICTED P(HIGH-BOOKING)

NO.	LISTING	P
01	Batignolles loft	59.2%
02	Cardinet studio	57.8%
03	Brochant 1-bed	56.4%
04	Pte. de Clichy flat	55.9%
05	Lemercier apartment	54.2%
06	La Fourche studio	53.1%
07	Guy-Môquet 1-bed	52.0%
08	Square des Batignolles	51.4%
09	Brochant courtyard	50.6%
10	Place de Lévis flat	49.3%

LISTING NAMES ANONYMISED BY METRO-STOP PROXIMITY.

CONCLUSION: INVESTOR RECOMMENDATION

Expected revenue net of vacancy. The aggressive option charges 60% more per night and earns less annually.

OPTION A	OPTION B	OPTION C
CONSERVATIVE	MODERATE	AGGRESSIVE
North · Batignolles	North · Batignolles	South · Ternes
PRICE / NIGHT	PRICE / NIGHT	PRICE / NIGHT
CANCELLATION	CANCELLATION	CANCELLATION
ROOM TYPE	ROOM TYPE	ROOM TYPE
P(HIGH BOOKING)	P(HIGH BOOKING)	P(HIGH BOOKING)
\$75	\$85	\$120
Flexible	Flexible	Strict
Entire apt.	Entire apt.	Entire apt.
16.2%	15.8%	9.2%



The aggressive scenario charges 60% more per night than the conservative and earns less annually. Booking probability collapses. This market does not reward premium pricing.

PART II · RECOMMENDATION	DRAWING P-10 OPTIONS	SCALE N.T.S.	SHEET
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